MIL-P-44206A
AMENDMENT-2
16 December 1987
SUPERSEDING
AMENDMENT-1
5 February 1987

MILITARY SPECIFICATION

POTATOES AU GRATIN, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This amendment forms a part of MIL-P-44206A, dated 18 April 1986, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, under "Other Government Documents": Add:

"U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Cheddar Cheese

Dairy Plants Surveyed and Approved for USDA Grading Service

(Application for copies should be addressed to Dairy Standardization Section, Dairy Division, Room 2750-So., Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)"

Under U.S. Department of Health and Human Services: Delete "U.S. Standards of Identity for Pasteurized Processed Cheese".

2.2: Add the following:

"AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)"

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* 3.2.1, line 5: Delete "3/8 by 3/8 by 1/2 inch dimensions" and substitute "3/8 inch cubes."

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3.2.2: Delete and substitute:

* "3.2.2 Cheese, cheddar, sharp, dehydrated, uncolored. Dehydrated cheddar cheese shall be derived from sharp, aged cheddar cheese, U.S. Grade A or better of the U.S. Standards for Grades of Cheddar Cheese. The dehydrated cheese shall be white in color, uniform in composition and appearance, and free from burnt specks and from lumps that do not fall apart under light finger pressure. The cheese shall possess a characteristic flavor and odor. The dehydrated cheese shall contain not more than 4.0 percent moisture, not more than 6.0 percent salt, and not less than 45.0 percent milkfat (see 6.3)."

3.2.3, line 2 after "aroma": Insert ".It".

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Add new ingredient paragraph:

* "3.2.11 Color, Annatto, dry. Annatto color shall be a reddish-orange water soluble powder of the extractive of Annatto bixa Orellana. The powder shall contain not more than 5.0 percent moisture and 100 percent of the powder shall pass through a U.S. Standard No. 10 sieve. The total color (as percent Norbixin) shall be 15 percent."

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3.3.1 Delete and substitute:

"3.3.1 Preparation of dehydro-frozen potatoes. The dehydro-frozen potato dices shall be blanched to approximately 1.5 times the frozen weight. The blanched potatoes shall be handled in such a manner as to prevent discoloration and shall be filled into the pouches within 4 hours after blanching."

3.3.2, lines 4 through 11: Delete and substitute:

*	"Water	80.963
	Cheese, cheddar, dehydrated	8.24
	Margarine	4.12
	Shortening, vegetable, powdered	3.64
	Starch, waxy maize, modified	2.11
	Salt 1/	0.52
	Onion powder	0.34
	Pepper, white	0.06
*	Annatto	0.007

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PAGE 5 (cont'd)

* 3.4b. Delete in entirety.

* 3.4c, line 1: Delete "1" and substitute "2".

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* 3.6f, after "2.25 ounces": Add "or more than 2.75 ounces."

3.6k, line 2: Delete "1.0" and substitute "0.5" and delete "1.5" and substitute "1.0".

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4.5.1.1: Delete and substitute:

"4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, US Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product."

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* 4.5.5 and table I: Delete in entirety and substitute the following new paragraphs and tables:

"4.5.5 Product and drained weight inspection.

4.5.5.1 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The pouches shall be heated in 150° to 160°F water for 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table I. The contents shall be inspected for the defects indicated in table II.

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TABLE I. Double sampling plan for product inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	Potato texture mushy, hard, fibrous, or tough.
102	Potato dice not approximately uniform in size and shape.
103	Sauce consistency lumpy or is excessively thick or thin.
104	Product shows evidence of excessive heating (materially darkened or scorched).

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

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4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65° to 75°F) after completion of the thermoprocessing operation. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the double sampling plan specified in table III. The drained weight shall be determined and inspected for the defect listed in table IV.

TABLE III. Double sampling plan for product inspection 1/ 2/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second sample shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

2/ To determine drained weight, the pouch contents shall be poured into a flat bottom container. A minimum of three times the pouch's volume of 180° to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated such as to separate sauce from potatoes. The contents shall then be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

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TABLE VI. Drained weight defect

Category	Defect
Major	
101	Drained weight of diced potatoes less than 2.0 ounces. 1/

1/ If the sample average drained weight is less than 2.25 ounces or more than 2.75 ounces, the lot shall be rejected.

Add the following new paragraph:

* "6.3 Cheese, cheddar, sharp, dehydrated. Kraft Cheeztang, spray dried, uncolored, product No. 9378 and Land O Lakes spray dried , white, product No. 49122 were found to be satisfactory in the development of this product."

Add the following new paragraph:

6.4 Changes from previous amendment. The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0626

Review activities:

Army - MD, TS
Navy - MC
DP - SS

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